

Disentangling the overlap between frailty and intrinsic capacity in older adults

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ABSTRACT

Introduction: It has been suggested that two prominent frameworks in geriatrics, frailty and intrinsic capacity (IC), represent two opposite ends of the same continuum. Frailty quantifies accumulated deficits and vulnerability, and IC measures an individual's capacities and functional reserves. This study investigates the overlap between frailty and IC in a large cohort of older adults.

Methods: We analysed 3246 participants aged 55+ from the Longitudinal Aging Study Amsterdam. Participants were categorised into four groups: neither frail nor low IC, low IC only, frail only, and both frail and low IC. Overlap was examined across age groups. Domain scores (vitality, sensory, cognition, psychology, locomotion) were compared between groups.

Results: Among 57–59 year olds, only 2.2 % were both frail and had low IC. This proportion increased with age, reaching 53.6 % at ages 87–89, while the “frail only” and “low IC only” groups declined. Overlap between frailty and domain scores showed that the “Neither” group consistently had the highest IC scores across all five domains, while the “Both” group consistently had the lowest scores.

Conclusions: In later life, being frail does not necessarily imply low IC, and vice versa. Frailty and low IC identify different groups in early older age, but the overlap between them becomes more pronounced with increasing age. Comprehensive assessment of ageing therefore requires measures that capture both vulnerability to decline and capacity.

1. Introduction

The promotion of healthy ageing requires effective ways to measure older adults' health status. Frailty is a state of vulnerability to poor resolution of homeostasis after stress, driven by accelerated loss of physiological reserve (Hoogendijk et al., 2019). It can be seen as a continuum from vital to frail, and is commonly operationalised through the accumulation of health deficits across multiple body systems (Bortz 2nd, 1993). While most research focuses on the process of becoming frail, it is equally important to recognise that many older adults remain relatively vital (Rockwood et al., 2011). To shift the focus beyond deficits, the World Health Organization recently introduced intrinsic capacity (IC), defined as the composite of an individual's physical and mental capacities (World Health Organization, 2017; Cesari et al.,

2018). IC aims to capture the positive attributes that support functional ability, including domains such as vitality, sensory function, cognition, psychological well-being, and locomotion. Whereas frailty reflects accumulated deficits, IC is a positive measure of functional reserves (Beyene et al., 2024; Belloni and Cesari, 2019). Both frailty and IC offer important but different perspectives on the ageing process and may inform complementary approaches to assessment and care.

The conceptual and empirical overlap between frailty and IC remains largely unknown. In this study, we used longitudinal data to describe overlap between these two constructs across age groups, rather than to model individual trajectories. To our knowledge, this age-specific perspective on the overlap between the two concepts has not been examined before.

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2. Methods

2.1. Study population

We used data from the Longitudinal Ageing Study Amsterdam (LASA), an ongoing nationally representative cohort of older adults in the Netherlands. LASA was approved by the medical ethics committee of the VU University Medical Center, and all participants provided written informed consent. Details on sampling and data collection have been published elsewhere (Hoogendijk et al., 2025). For this study, we analysed 3246 participants aged 55 and older, with follow-up at 3, 6, and 9 years.

2.2. Measures

2.2.1. Intrinsic capacity

The IC measure used in this study was previously developed and validated within the LASA cohort (Qi et al., 2024). Domain-specific scores were calculated for the five IC domains: vitality, sensory, cognition, psychology, and locomotion. Specifically, vitality was measured using hand grip strength (George et al., 2021), peak expiratory flow rate (George et al., 2021; van Schoor et al., 2012; Beard et al., 2022), and calf circumference (Sanchez-Rodriguez et al., 2023). The sensory domain included self-rated vision and hearing items (Gutiérrez-Robledo et al., 2019). Cognition was assessed via the Mini-Mental State Examination (López-Ortiz et al., 2022), verbal memory task (Koivunen et al., 2023), and letter coding task (Koivunen et al., 2023). Psychology included anxiety (López-Ortiz et al., 2022), Center for Epidemiologic Studies Depression Scale, mastery (Golino et al., 2020), self-efficacy (Koivunen et al., 2023), and self-esteem (Astrone et al., 2022). Locomotion was measured using walking speed (George et al., 2021; Chen et al., 2023), chair rise (Beard et al., 2019), and cardigan tests. A composite IC score was calculated as the average of the five domain scores. To enable comparison across studies, all scores were rescaled using the Proportion of Maximum Possible (POMP) method (Cohen et al., 2013; Cohen et al., 1999), with domain and composite scores ranging from 0 (low capacity) to 100 (high capacity).

2.2.2. Frailty

In LASA, a 32-item frailty index was previously developed and validated (Hoogendijk et al., 2017). A overview of included items is available in the validation study (Hoogendijk et al., 2017). All items were scored as 0 (deficit absent) or 1 (deficit present), and the total score was calculated by dividing the sum of deficits by the number of items, yielding a value from 0 to 1. To allow comparison with IC, the score was inverted ($1 - \text{frailty index}$) and scaled using the Proportion of Maximum Possible (POMP) method, resulting in a final score from 0 (frail) to 100 (not frail). An overview of the variables included in the frailty index can be found in Appendix 1.

2.3. Statistical analysis

For this analysis, participant data were grouped into 3-year age intervals (e.g. 57–59, 60–62, up to 90+) based on their age at each measurement wave. This allowed us to examine how frailty and IC classifications varied by age rather than by measurement wave. Participants were categorised into four mutually exclusive groups based on frailty and IC status. Low IC was defined as a composite IC score in the lowest quartile of the total sample. Frailty was defined as a reversed frailty score ≤ 75 , equivalent to an original frailty index ≥ 0.25 , a commonly used threshold (Hoogendijk et al., 2020a). The four resulting groups were: individuals who were neither frail nor had low IC; those with low IC only; those who were frail only; and those who were both frail and had low IC. For each age group, we calculated the proportion of participants in each category. We also computed the median scores for each of the five IC domains (vitality, sensory, cognition, psychology, and

locomotion) within each of the four status groups.

3. Results

Fig. 1 shows, for each age group, the proportions classified as neither, low IC only, frail only, or both. In 57–59 years ($N = 179$), most individuals were neither frail nor had low IC (87.8 %), with smaller proportions in “frail only” (7.3 %), “low IC only” (2.8 %), and “both” (2.2 %). With increasing age, the share of individuals in the “neither” group declined. By 87–89 years ($N = 239$), “both” was the largest group (53.6 %), and in 90+ the distribution was: neither 8.6 %, low IC only 29.6 %, frail only 4.9 %, and both 56.8 %. The percentage of older adults with low IC increased with age, while the frail-only group remained relatively stable (around 7 %) across most ages, with a lower proportion in 90+ (4.9 %).

Fig. 2 shows median scores for vitality, sensory, cognition, psychology, and locomotion by age group and combined frailty and intrinsic capacity status. Across all domains and age groups, individuals who were neither frail nor had low IC consistently had the highest scores, while those with both conditions had the lowest. In most domains, individuals who were frail only scored higher than those with low IC only.

4. Discussion

This study aimed to describe the overlap between frailty and IC during ageing using data from a large longitudinal cohort of older adults. We examined how the distribution of individuals classified as frail, low IC, both, or neither changed with age. The results show that frailty and low IC identify different groups, especially at younger ages. Being frail does not necessarily imply low IC, and vice versa. With age, the proportion classified as both increases, while the share with neither declines. Median scores across five IC domains reveal that individuals with both conditions consistently had the lowest scores, and those with neither had the highest.

Frailty and low IC do not identify the same group of individuals in the early stages of ageing. Frailty measures vulnerability while IC quantifies functional reserves (Belloni and Cesari, 2019). As individuals age, deficits may accumulate and reserves decline, leading to greater overlap between the two. However, individuals with either frailty or low IC alone remain present in older age, even though the measures partly share the same items. This suggests that screening for IC remains relevant alongside frailty, perhaps as an indicator of resilience. For example, someone with low IC but not yet frail may be at risk for future decline before meeting frailty criteria. Chew et al. (Chew et al., 2021) used cluster analysis to identify subgroups based on frailty and IC domains, showing that IC may complement frailty in relatively healthy older adults and support risk stratification. These findings reinforce that frailty and IC are distinct constructs. Both are relevant for assessing health in older adults, but it remains unclear when each should be used, whether they offer added value when combined, and whether their clinical relevance varies by age. IC is still rarely used in practice, and further research is needed to determine whether frailty and IC should be assessed simultaneously or whether low IC predicts future frailty (or vice versa). Longitudinal studies on transitions between these states are needed to clarify timing and use in clinical care. Although we cannot yet provide definitive recommendations for the clinical use of frailty and IC, as further evidence is still needed, a stepwise approach may offer a practical solution. For instance, initial brief indicators such as walking speed (Koivunen et al., 2024) or fatigue could be used to flag individuals for further assessment. Once we better understand whether frailty or IC decline tends to precede the other, we can offer more targeted guidance on when and how to assess each construct to maximise both clinical benefit and efficiency.

To support future research and clinical application, it is also important to consider how frailty is measured. The frailty index used in this study captures a broad range of age-related health deficits and has

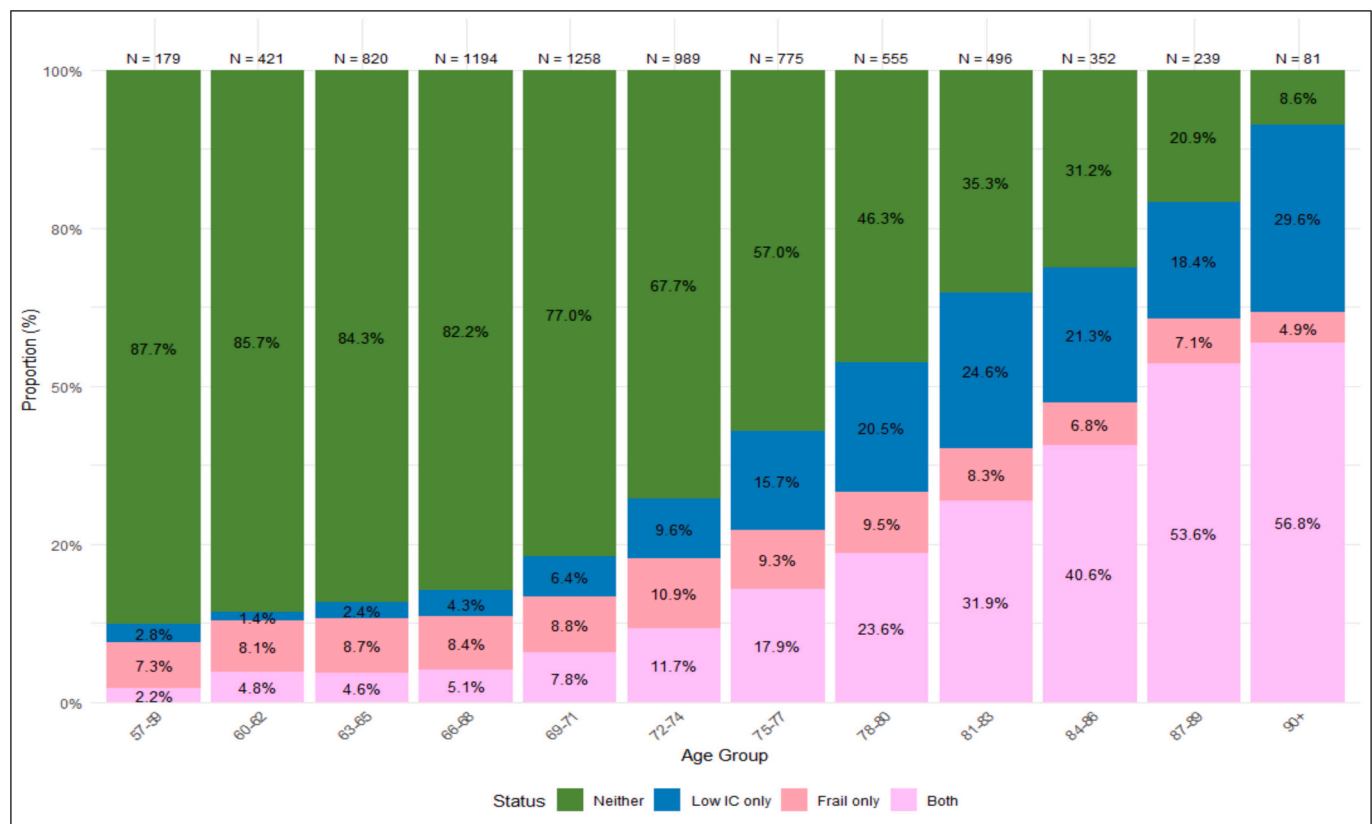


Fig. 1. Proportion of older adults classified by frailty and low intrinsic capacity across age groups.

shown to have good validity and sensitivity to changes during ageing (de Breij et al., 2021; Stolz et al., 2020; Hoogendijk et al., 2021). However, in LASA, several other frailty measures have also been operationalised, each reflecting different conceptualisations. These include the Fried phenotype, which focuses on physical frailty using five indicators (e.g. weight loss, exhaustion, low physical activity) (Fried et al., 2021; Hoogendijk et al., 2020b); and the functional domains approach, which groups deficits across physical, nutritive, cognitive, and sensory domains (Hoogendijk et al., 2018). While these tools offer valuable perspectives, operationalising them with our available data would require a greater reliance on self-reported indicators. Comparing frailty definitions and their alignment with IC could be an informative next step toward integrated assessment strategies.

The distinct IC domain profiles observed across frailty and IC groups provide insight into their clinical relevance and point to the need for a more nuanced approach to ageing assessment. Rather than focusing solely on an overall status such as frailty, assessing specific domains of decline could support the development of more targeted interventions (World Health Organization, 2017). This may be particularly relevant given that IC is still being refined, and relying on a single summary score may not yet capture the full complexity of ageing (Hoogendijk et al., 2023). Environmental factors also influence IC; for example, using glasses or hearing aids may help maintain capacity in the sensory domain and prevent declines that might otherwise be classified as low IC. The finding that individuals who were frail only often had higher domain scores than those with low IC only perhaps suggests that IC may detect earlier decline. In practice, this would help clinicians identify individuals at risk who are not yet frail but may benefit from preventive strategies, reducing the need for more comprehensive assessments later. However, further research is needed to determine how these scores can be best used to guide clinical decisions.

This exploratory study describes age-specific overlap between frailty

and IC, an area with limited prior evidence; however, several limitations should be considered. Although the use of a large longitudinal cohort allows analysis across age groups, the descriptive design does not permit causal inference. Second, as the concept of IC is still evolving, there is no universally accepted standard for its measurement, complicating comparisons across studies (World Health O, 2024). This also makes it difficult to define a clear threshold for low IC. In this study, we used the lowest quartile of the composite score, a choice that warrants further evaluation. Finally, while the cohort is representative of older adults in the Netherlands, generalisability to more diverse populations may be limited.

In conclusion, our results show that frailty and IC are distinct concepts that capture different aspects of ageing. Although they overlap more in advanced age, they continue to identify separate groups. These findings suggest that using both measures may help provide a complete picture of an individual's health status, but further research is needed to clarify when and how each measure is most useful in clinical practice.

CRediT authorship contribution statement

Yuwei Qi: Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Natasja M. van Schoor:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Laura A. Schaap:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Emiel O. Hoogendijk:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization.

Declaration of competing interest

None declared.

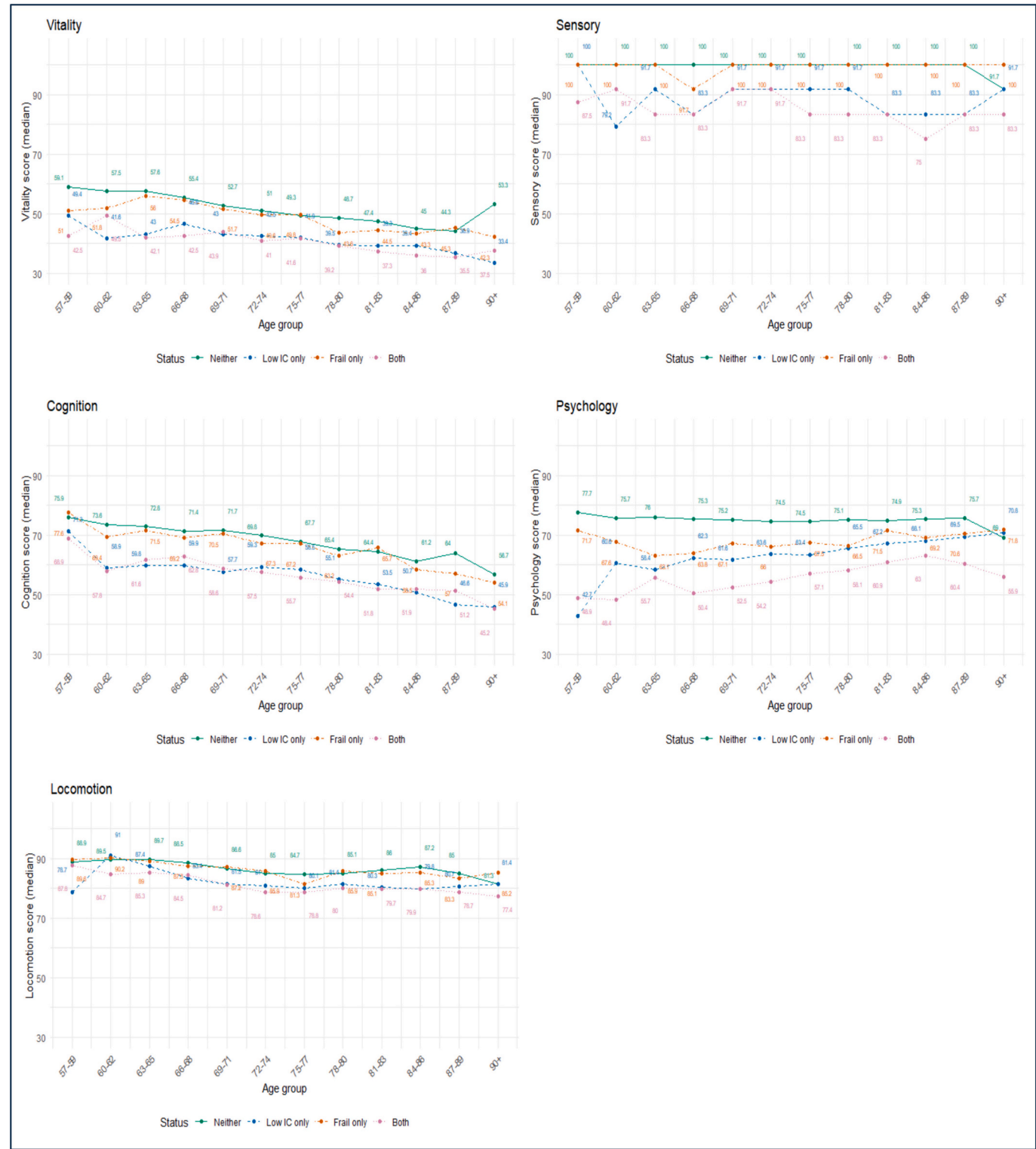


Fig. 2. Profiles of intrinsic capacity domains by combined frailty and intrinsic capacity status.

Appendix 1. Overview of the variables included in the frailty index

No.	Deficit	Cutoff values
1	Cardiac disease	No = 0, yes = 1
2	Peripheral arterial disease	No = 0, yes = 1
3	Stroke	No = 0, yes = 1
4	Diabetes	No = 0, yes = 1

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No.	Deficit	Cutoff values
5	Lung disease	No = 0, yes = 1
6	Cancer	No = 0, yes = 1
7	Arthritis	No = 0, yes = 1
8	Hypertension	No = 0, yes = 1
9	Other chronic disease 1	No = 0, yes = 1
10	Other chronic disease 2	No = 0, yes = 1
11	Incontinence	No = 0, yes = 1
12	Walk up/down staircase 15 steps without resting	Yes = 0, yes, with some difficulty = 0.25, yes, with much difficulty = 0.50, only with help = 0.75, No = 1
13	Dress/undress self	Yes = 0, yes, with some difficulty = 0.25, yes, with much difficulty = 0.50, only with help = 0.75, no = 1
14	Sit down/stand up from chair	Yes = 0, yes, with some difficulty = 0.25, yes, with much difficulty = 0.50, only with help = 0.75, no = 1
15	Cut own toenails	Yes = 0, yes, with some difficulty = 0.25, yes, with much difficulty = 0.50, only with help = 0.75, no = 1
16	Walk outside 5 min without stopping	Yes = 0, yes, with some difficulty = 0.25, yes, with much difficulty = 0.50, only with help = 0.75, no = 1
17	Use of transportation	Yes = 0, yes, with some difficulty = 0.25, yes, with much difficulty = 0.50, only with help = 0.75, no = 1
18	How is your health in general?	Excellent = 0, good = 0.25, fair = 0.50, sometimes good/bad = 0.75, Poor = 1
19	How is your health compared to other people of your age?	Much better/a little better = 0, just as good = 0.33, a little worse = 0.66, much worse = 1
20	Feel depressed (CES-D)	Rarely or never = 0, some of the time = 0.33, occasionally = 0.66, mostly or always = 1
21	Feel everything is an effort (CES-D)	Rarely or never = 0, some of the time = 0.33, occasionally = 0.66, mostly or always = 1
22	Feel happy (CES-D)	Mostly or always = 0, occasionally = 0.33, some of the time = 0.66, rarely or never = 1
23	Feel lonely (CES-D)	Rarely or never = 0, some of the time = 0.33, occasionally = 0.66, mostly or always = 1
24	Enjoy life (CES-D)	Mostly or always = 0, occasionally = 0.33, some of the time = 0.66, rarely or never = 1
25	Could not get going (CES-D)	Rarely or never = 0, some of the time = 0.33, occasionally = 0.66, mostly or always = 1
26	Physical activity (LAPAQ)	High (five or more activities) = 0, medium (3–4 activities) = 0.33, low (1–2 activities) = 0.66, no activities = 1
27	Memory complaints	No = 0, yes = 1
28	Orientation time (MMSE)	Five correct = 0, one wrong = 0.50, two or more wrong = 1
29	Orientation place (MMSE)	Five correct = 0, one wrong = 0.50, two or more wrong = 1
30	Attention (MMSE)	Five correct = 0, one or two wrong = 0.50, three or more wrong = 1
31	Recall (MMSE)	Three correct = 0, two correct = 0.50, one or zero correct = 1
32	Gait speed (to walk 3 m, turn around and walk back 3 m)	Normal = 0, slow (>10 s) or physical unable = 1

Data availability

The authors do not have permission to share data.

References

Astrone, P., Perracini, M.R., Martin, F.C., Marsh, D.R., Cesari, M., 2022. The potential of assessment based on the WHO framework of intrinsic capacity in fragility fracture prevention. *Aging Clin. Exp. Res.* 34 (11), 2635–2643.

Beard, J.R., Jotheeswaran, A.T., Cesari, M., Araujo de Carvalho, I., 2019. The structure and predictive value of intrinsic capacity in a longitudinal study of ageing. *BMJ Open* 9 (11), e026119.

Beard, J.R., Si, Y., Liu, Z., Chenoweth, L., Hanewald, K., 2022. Intrinsic capacity: validation of a new WHO concept for healthy aging in a longitudinal Chinese study. *J. Gerontol. A Biol. Sci. Med. Sci.* 77 (1), 94–100.

Belloni, G., Cesari, M., 2019. Frailty and intrinsic capacity: two distinct but related constructs. *Front. Med. (Lausanne)* 6, 133.

Beyene, M.B., Visvanathan, R., Amare, A.T., 2024. Intrinsic capacity and its biological basis: a scoping review. *J. Frailty Aging* 13 (3), 193–202.

Bortz 2nd, W.M., 1993. The physics of frailty. *J. Am. Geriatr. Soc.* 41 (9), 1004–1008.

Cesari, M., Araujo de Carvalho, I., Amuthavalli, T., Jotheeswaran, Cooper C., Martin, F. C., et al., 2018. Evidence for the domains supporting the construct of intrinsic capacity. *J. Gerontol.: Series A* 73 (12), 1653–1660.

Chen, Y.-J., Kukreti, S., Yang, H.-L., Liu, C.-C., Yeh, Y.-C., Fung, X.C.C., et al., 2023. Psychometric properties of instruments assessing intrinsic capacity: a systematic review. *Asian J. Soc. Health Behav.* 6 (4).

Chew, J., Lim, J.P., Yew, S., Yeo, A., Ismail, N.H., Ding, Y.Y., et al., 2021. Disentangling the relationship between frailty and intrinsic capacity in healthy community-dwelling older adults: a cluster analysis. *J. Nutr. Health Aging* 25 (9), 1112–1118.

Cohen, J., Cohen, P., West, S.G., Aiken, L.S., 2013. *Applied Multiple Regression/Correlation Analysis for the Behavioral Sciences*. Routledge.

Cohen, P., Cohen, J., Aiken, L.S., West, S.G., 1999. The problem of units and the circumstance for POMP. *Multivar. Behav. Res.* 34 (3), 315–346.

de Breijl, S., van Hout, H.P.J., de Bruin, S.R., Schuster, N.A., Deeg, D.J.H., Huisman, M., et al., 2021. Predictors of frailty and vitality in older adults aged 75 years and over: results from the longitudinal aging study Amsterdam. *Gerontology* 67 (1), 69–77.

Fried, L.P., Cohen, A.A., Xue, Q.-L., Walston, J., Bandeen-Roche, K., Varadhan, R., 2021. The physical frailty syndrome as a transition from homeostatic symphony to cacophony. *Nat. Aging* 1 (1), 36–46.

George, P.P., Lun, P., Ong, S.P., Lim, W.S., 2021. A rapid review of the measurement of intrinsic capacity in older adults. *J. Nutr. Health Aging* 25 (6), 774–782.

Golino, H., Thiagarajan, J.A., Sadana, R., Teles, M., Christensen, A.P., Boker, S.M., 2020. Investigating the Broad Domains of Intrinsic Capacity, Functional Ability and Environment: An Exploratory Graph Analysis Approach for Improving Analytical Methodologies for Measuring Healthy Aging.

Gutiérrez-Robledo, L.M., García-Chanes, R., Pérez-Zepeda, M., 2019. Allostatic load as a biological substrate to intrinsic capacity: a secondary analysis of CRELES. *J. Nutr. Health Aging* 23 (9), 788–795.

Hoogendijk, E.O., Theou, O., Rockwood, K., Onwuteaka-Philipsen, B.D., Deeg, D.J.H., Huisman, M., 2017. Development and validation of a frailty index in the longitudinal aging study Amsterdam. *Aging Clin. Exp. Res.* 29 (5), 927–933.

Hoogendijk, E.O., Heymans, M.W., Deeg, D.J.H., Huisman, M., 2018. Socioeconomic inequalities in frailty among older adults: results from a 10-year longitudinal study in the Netherlands. *Gerontology* 64 (2), 157–164.

Hoogendijk, E.O., Afilalo, J., Ensrud, K.E., Kowal, P., Onder, G., Fried, L.P., 2019. Frailty: implications for clinical practice and public health. *Lancet* 394 (10206), 1365–1375.

Hoogendijk, E.O., Stenholm, S., Ferrucci, L., Bandinelli, S., Inzitari, M., Cesari, M., 2020a. Operationalization of a frailty index among older adults in the InCHIANTI study: predictive ability for all-cause and cardiovascular disease mortality. *Aging Clin. Exp. Res.* 32 (6), 1025–1034.

Hoogendijk, E.O., Smit, A.P., van Dam, C., Schuster, N.A., de Breijl, S., Holwerda, T.J., et al., 2020b. Frailty combined with loneliness or social isolation: an elevated risk for mortality in later life. *J. Am. Geriatr. Soc.* 68 (11), 2587–2593.

Hoogendijk, E.O., Stolz, E., Oude Voshaar, R.C., Deeg, D.J.H., Huisman, M., Jeurings, H. W., 2021. Trends in frailty and its association with mortality: results from the longitudinal aging study Amsterdam, 1995–2016. *Am. J. Epidemiol.* 190 (7), 1316–1323.

Hoogendijk, E.O., Dent, E., Koivunen, K., 2023. Intrinsic capacity: an under-researched concept in geriatrics. *Age Ageing* 52, afad183.

Hoogendijk, E.O., van Schoor, N.M., Qi, Y., Visser, M., Swinkels, J.C., Broese van Groenou, M.I., et al., 2025. The longitudinal aging study Amsterdam: design and cohort update 2025. *Eur. J. Epidemiol.* 1–16.

Koivunen, K., Hoogendijk, E.O., Schaap, L.A., Huisman, M., Heymans, M.W., van Schoor, N.M., 2023. Development and validation of an intrinsic capacity composite score in the longitudinal aging study Amsterdam: a formative approach. *Aging Clin. Exp. Res.* 35 (4), 815–825.

Koivunen, K., Lindeman, K., Välimaa, M., Rantanen, T., 2024. Investigating Resilience Through Intrinsic Capacity Networks in Older Adults. *J. Gerontol. A. Biol. Sci. Med. Sci.* 79 (10), glae048. <https://doi.org/10.1093/gerona/glae048>.

López-Ortiz, S., Lista, S., Peñín-Grandes, S., Pinto-Fraga, J., Valenzuela, P.L., Nisticò, R., et al., 2022. Defining and assessing intrinsic capacity in older people: a systematic review and a proposed scoring system. *Ageing Res. Rev.* 79, 101640.

Qi, Y., Schaap, L.A., Schalet, B.D., Hoogendijk, E.O., Deeg, D.J.H., Visser, M., et al., 2024. The development of intrinsic capacity measures for longitudinal research: the longitudinal ageing study Amsterdam. *Exp. Gerontol.* 197, 112599. <https://doi.org/10.1016/j.exger.2024.112599>.

Rockwood, K., Song, X., Mitnitski, A., 2011. Changes in relative fitness and frailty across the adult lifespan: evidence from the Canadian National Population Health Survey. *Cmaj* 183 (8), E487–E494.

Sanchez-Rodriguez, D., Demonceau, C., Bruyère, O., Cavalier, E., Reginster, J.Y., Beaudart, C., 2023. Intrinsic capacity and risk of death: focus on the impact of using different diagnostic criteria for the nutritional domain. *Maturitas* 176, 107817.

- Stolz, E., Hoogendijk, E.O., Mayerl, H., Freidl, W., 2020. Frailty changes predict mortality in 4 longitudinal studies of aging. *J. Gerontol.: Series A*. 76 (9), 1619–1626.
- van Schoor, N.M., de Jongh, R.T., Daniels, J.M., Heymans, M.W., Deeg, D.J., Lips, P., 2012. Peak expiratory flow rate shows a gender-specific association with vitamin D deficiency. *J. Clin. Endocrinol. Metab.* 97 (6), 2164–2171.
- World Health O, 2024. Measuring the Progress and Impact of the UN Decade of Healthy Ageing (2021–2030): Framework and Indicators Recommended by WHO Technical Advisory Group. World Health Organization, Geneva, 2024–12–31.
- World Health Organization, 2017. WHO clinical consortium on healthy ageing: topic focus: frailty and intrinsic capacity: report of consortium meeting, 1–2 December 2016 in Geneva, 2017. World Health Organization, Switzerland. Geneva. Contract No.: WHO/FWC/ALC/17.2.